Optimal approximate-counting bounds with trusted quantum samples, reflections and a QSampling unitary

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Editorial note. Editable transcription of corresponding formally published CC BY3.0 Unported trusted-state counting result. Nonexclusive nativev3 remains private; public selected files exclude all unselected raw ranges and comments. Full source-local Sections4.1–4.4/4.6 and AppendixA preserved, matched algorithms and specific-unitary result combined. Unused classical-sample sketch4.5 and later discussion/Followup section excluded. Original author probabilities, parameter choices and equations retained without mathematical repair or added assumptions. Original theorem/section numbering restored across omitted classicalPropositions29/30. Identity glyph uses ordinary bold1 instead of unavailable custom bbold font; no mathematical change. Bibliography identities matched to formal numbered entries. All production/validation onlyYupeng; root owns compilation.

Allowed standard prerequisites (editorial).

- Quantum query algorithms and QMA amplification; approximation theory and symmetric polynomial symmetrization. Full originalSection2 definitions/statements, source-local symmetrization proofs and guessing lemma retained; standard cited amplification is not recursively reproved

Trusted quantum samples and reflections: model and promises

In the $\mathsf{ApxCount}_{N,w}$ problem, our goal is to decide whether a nonempty set $S \subseteq [N]$ satisfies $|S| \geq 2w$ (YES) or $|S| \leq w$ (NO), promised that one of these is the case. In the query model the algorithm is given a membership oracle for S. QSampling S means that we can approximately prepare the uniform superposition

$$|S\rangle := \frac{1}{\sqrt{|S|}} \sum_{i \in S} |i\rangle.$$

To be clear, QSampling supposes that the algorithm is given trusted copies of $|S\rangle$; unlike in the QMA setting, the state need not be "verified" by the algorithm. Reflections about $|S\rangle$ apply the unitary transformation

$$\mathcal{R}_S := \mathbf{1} - 2|S\rangle\langle S|,$$

which has eigenvalue -1 for $|S\rangle$ and eigenvalue $+1$ for all states orthogonal to $|S\rangle$. More concretely, let U be the unitary that performs the map $U|0\rangle = |S\rangle$, for some canonical starting state $|0\rangle$. Since we know the circuit U, we can also implement $U^\dagger$, by reversing the order of all the gates and replacing all the gates with their adjoints. Then

$$\mathcal{R}_S = \mathbf{1} - 2|S\rangle\langle S| = U(\mathbf{1} - 2|0\rangle\langle 0|)U^\dagger.$$

We begin by focusing on the slightly simplified setting where the algorithm is only permitted to perform membership queries, QSamples, and reflections about the state $|S\rangle$. The lower bound below even holds for the exact-size promise $|S| = w$ versus $|S| = 2w$; the full unitary model is specified in original Section4.6, retained below.

Theorem 4. *Let Q be a quantum algorithm that makes T queries to the membership oracle for S, and uses a total of R copies of $|S\rangle$ and reflections about $|S\rangle$. If Q decides whether $|S| = w$ or $|S| = 2w$ with high probability, promised that one of those is the case, then either*

$$T = \Omega\left(\sqrt{\frac{N}{w}}\right) \qquad or \qquad R = \Omega\left(\min\left\{w^{1/3}, \sqrt{\frac{N}{w}}\right\}\right). \tag{1}$$

Theorem 5. *There is a quantum algorithm that solves $\mathsf{ApxCount}_{N,w}$ with high probability using R copies of $|S\rangle$ and reflections about $|S\rangle$, where $R = O\left(\min\left\{w^{1/3}, \sqrt{\frac{N}{w}}\right\}\right)$.*

2 Preliminaries

In this section we introduce some definitions and known facts about polynomials and complexity classes.

2.1 Approximation theory

We will use several results from approximation theory, each of which has previously been used (in some form) in other applications of the polynomial method to quantum lower bounds. We start with the basic inequality of A.A. Markov [35].

Lemma 6 (Markov)**.** *Let p be a real polynomial, and suppose that*

$$\max_{x,y\in[a,b]} |p(x) - p(y)| \leq H. \tag{5}$$

Then for all $x \in [a,b]$, we have

$$|p'(x)| \leq \frac{H}{b-a}\deg(p)^2, \tag{6}$$

where $p'(x)$ is the derivative of p at x.

We'll also need a bound that was explicitly stated by Paturi [43], and which amounts to the fact that, among all degree-d polynomials that are bounded within a given range, the Chebyshev polynomials have the fastest growth outside that range.

Lemma 7 (Paturi)**.** *Let p be a real polynomial, and suppose that $|p(x)| \leq 1$ for all $|x| \leq 1$. Then for all $x \leq 1+\mu$, we have*

$$|p(x)| \leq \exp\left(2\deg(p)\sqrt{2\mu+\mu^2}\right). \tag{7}$$

We now state a useful corollary of Lemma 7, which says (in effect) that slightly shrinking the domain of a low-degree real polynomial can only modestly shrink its range.

Corollary 8. *Let p be a real polynomial of degree d, and suppose that*

$$\max_{x,y\in[a,b]} |p(x)-p(y)| \geq H. \tag{8}$$

Let $\varepsilon \leq \frac{1}{100d^2}$ and $a' := a+\varepsilon(b-a)$. Then

$$\max_{x,y\in[a',b]} |p(x)-p(y)| \geq \frac{H}{2}. \tag{9}$$

Proof. Suppose by contradiction that

$$|p(x)-p(y)| < \frac{H}{2} \tag{10}$$

for all $x, y \in [a', b]$. By affine shifts, we can assume without loss of generality that $|p(x)| < \frac{H}{4}$ for all $x \in [a', b]$. Then by Lemma 7, for all $x \in [a, b]$ we have

$$|p(x)| < \frac{H}{4} \cdot \exp\left(2d\sqrt{2\left(\frac{1}{1-\varepsilon}-1\right)+\left(\frac{1}{1-\varepsilon}-1\right)^2}\right) \leq \frac{H}{2}. \tag{11}$$

But this violates the hypothesis. □

We will also need a bound that relates the range of a low-degree polynomial on a discrete set of points to its range on a continuous interval. The following lemma generalizes a result due to Ehlich and Zeller [22] and Rivlin and Cheney [46], who were interested only in the case where the discrete points are evenly spaced.

Lemma 9. *Let p be a real polynomial of degree at most $\sqrt{k}$, and let $0 = z_1 < \cdots < z_M = k$ be a list of points such that $z_{i+1} - z_i \leq 1$ for all i (the simplest example being the integers $0, \ldots, k$). Suppose that*

$$\max_{x,y\in[0,k]} |p(x)-p(y)| \geq H. \tag{12}$$

Then

$$\max_{i,j} |p(z_i)-p(z_j)| \geq \frac{H}{2}. \tag{13}$$

Proof. Suppose by contradiction that

$$|p(z_i) - p(z_j)| < \frac{H}{2} \tag{14}$$

for all i, j. By affine shifts, we can assume without loss of generality that $|p(z_i)| < \frac{H}{4}$ for all i. Let

$$c := \max_{x \in [0,k]} \frac{|p(x)|}{H/4}. \tag{15}$$

If $c \leq 1$, then the hypothesis clearly fails, so assume $c > 1$. Suppose that the maximum, $|p(x)| = \frac{cH}{4}$, is achieved between z_i and z_{i+1}. Then by basic calculus, there exists an $x^* \in [z_i, z_{i+1}]$ such that

$$|p'(x^*)| > \frac{2(c-1)}{z_{i+1} - z_i} \cdot \frac{H}{4} \geq \frac{(c-1)H}{2}. \tag{16}$$

So by Lemma 6,

$$\frac{(c-1)H}{2} < \frac{cH/4}{k} \deg(p)^2. \tag{17}$$

Solving for c, we find

$$c < \frac{2k}{2k - \deg(p)^2} \leq 2. \tag{18}$$

But if $c < 2$, then $\max_{x \in [0,k]} |p(x)| < \frac{H}{2}$, which violates the hypothesis. □

We also use a related inequality due to Coppersmith and Rivlin [20] that bounds a polynomial on a continuous interval in terms of a bound on a discrete set of points, but now with the weaker assumption that the degree is at most k, rather than $\sqrt{k}$. This gives a substantially weaker bound.

Lemma 10 (Coppersmith and Rivlin)**.** *Let p be a real polynomial of degree at most k, and suppose that $|p(x)| \leq 1$ for all integers $x \in \{0, 1, \ldots, k\}$. Then there exist universal constants a, b such that for all $x \in [0, k]$, we have*

$$|p(x)| \leq a \cdot \exp\left(b \deg(p)^2/k\right). \tag{19}$$

2.2 Symmetric polynomials

Univariate symmetrizations. Our starting point is the well-known *symmetrization lemma* of Minsky and Papert [39] (see also Beals et al. [8] for its application to quantum query complexity), by which we can often reduce questions about multivariate polynomials to questions about univariate ones.

Lemma 11 (Minsky–Papert symmetrization)**.** *Let $p : \{0,1\}^N \to \mathbb{R}$ be a real multilinear polynomial of degree d, and let $q : \{0, 1, \ldots, N\} \to \mathbb{R}$ be defined as*

$$q(k) := \mathbb{E}_{|X|=k}[p(X)]. \tag{20}$$

Then q can be written as a real polynomial in k of degree at most d.

We now introduce a different, lesser known notion of symmetrization, which we call the *erase-all-subscripts* symmetrization for reasons to be explained shortly. This symmetrization previously appeared in [51] under the name "linearization," and it is also equivalent to the noise operator used in analysis of Boolean functions [42, Definition 2.46].

Lemma 12 (Erase-all-subscripts symmetrization)**.** *Let $p : \{0,1\}^N \to \mathbb{R}$ be a real multilinear polynomial of degree d, and for any real number $k \in [0,1]$, let M_k denote the distribution over $\{0,1\}^N$, wherein each coordinate is selected independently to be 1 with probability k. Let $q : [0,1] \to \mathbb{R}$ be defined as*

$$q(k) := \mathbb{E}_{X \sim M_k}[p(X)]. \tag{21}$$

Then q can be written as a real polynomial in k of degree at most d.

Proof. (see, for example, [47, Proof of Theorem 3]). Given the multivariate polynomial expansion of p, we can obtain q easily just by "erasing all the subscripts in each variable". For example, if $p(x_1, x_2, x_3) = 2x_1x_2 + x_2x_3 + x_2$, we replace every x_i with k to obtain $q(k) = 2k \cdot k + k \cdot k + k = 3k^2 + k$. This follows from linearity of expectation along with the fact that M_k is defined to be the product distribution wherein each coordinate has expected value k. □

We highlight the following key difference between Minsky–Papert symmetrization and the erase-all-subscripts symmetrization. Let $p : \{0,1\}^N \to [0,1]$ be a real multivariate polynomial whose evaluations at Boolean inputs are in $[0,1]$, i.e., for all $x \in \{0,1\}^n$, we have $p(x) \in [0,1]$. If q is the erase-all-subscripts symmetrization of p, then q takes values in $[0,1]$ at all *real-valued* inputs in $[0,1]$: $q(k) \in [0,1]$ for all $k \in [0,1]$. If q is the Minsky–Papert symmetrization of p, then it is only guaranteed to take values in $[0,1]$ at *integer-valued* inputs in $[0,N]$, i.e., $q(k) \in [0,1]$ is only guaranteed to hold at $k \in \{0,1,\ldots,N\}$. This is the main reason we use erase-all-subscripts symmetrization in this work.

Bivariate symmetrizations. In this paper, it will be convenient to consider bivariate versions of both Minsky–Papert and erase-all-subscripts symmetrization, and their applications to oracle separations. To this end, define $X \in \{0,1\}^N$, the "characteristic string" of the set $S \subseteq [N]$, by $x_i = 1$ if $i \in S$ and $x_i = 0$ otherwise. Let $\mathcal{O}_S$ denote the unitary that performs a membership query to S, defined as

$$\mathcal{O}_S |i\rangle |b\rangle = (1 - 2bx_i) |i\rangle |b\rangle \tag{22}$$

for any index $i \in [N]$ and bit $b \in \{0,1\}$.

Because we study oracle intersection problems, it is often convenient to think of an algorithm as having access to *two* oracles, wherein the first bit in the oracle register selects the choice of oracle. As a consequence, we need a slight generalization of a now well-established fact in quantum complexity: that the acceptance probability of a quantum algorithm with an oracle can be expressed as a polynomial in the bits of the oracle string.

Lemma 13 (Symmetrization with two oracles)**.** *Let $Q^{\mathcal{O}_{S_0}, \mathcal{O}_{S_1}}$ be a quantum algorithm that makes T queries to a pair of membership oracles for sets $S_0, S_1 \subseteq [N]$. Let D_μ denote the distribution over subsets of $[N]$ wherein each element is selected independently with probability $\frac{\mu}{N}$. Then there exist bivariate real polynomials $q(s,t)$ and $p(x,y)$ of degree at most $2T$ satisfying:*

$$\textit{for all real numbers } s,t \in [0,N], \quad q(s,t) = \mathbb{E}_{\substack{S_0 \sim D_s, \\ S_1 \sim D_t}} \left[\Pr[Q^{\mathcal{O}_{S_0}, \mathcal{O}_{S_1}} \textit{ accepts}]\right], \textit{ and}$$

$$\textit{for all integers } x,y \in \{0,1,\ldots,N\}, \quad p(x,y) = \mathbb{E}_{\substack{|S_0|=x, \\ |S_1|=y}} \left[\Pr[Q^{\mathcal{O}_{S_0}, \mathcal{O}_{S_1}} \textit{ accepts}]\right].$$

Proof. Take $X = X_0|X_1$ to be the concatenation of the characteristic strings of the two oracles, and let $S \subseteq [2N]$ be such that X is the characteristic string of S. Then, Lemma 4.2 of Beals et al. [8] tells us that there is a real multilinear polynomial $r(X)$ of degree at most $2T$ in the bits of X such that $r(X) = \Pr[Q^{\mathcal{O}_S} \text{ accepts}]$.

Observe that r has a meaningful probabilistic interpretation over arbitrary inputs in $[0,1]$. A vector $X \in [0,1]^{2N}$ of probabilities corresponds to a distribution over $\{0,1\}^{2N}$ wherein each bit is chosen from a Bernoulli distribution with the corresponding probability. Because r is multilinear, r in fact computes the expectation of the acceptance probability over this distribution. In particular, the polynomial

$$q(s,t) = r\Bigg(\underbrace{\frac{s}{N},\ldots,\frac{s}{N}}_{N \text{ times}},\underbrace{\frac{t}{N},\ldots,\frac{t}{N}}_{N \text{ times}}\Bigg) = \mathbb{E}_{\substack{S_0 \sim D_s,\\ S_1 \sim D_t}}\left[\Pr[Q^{\mathcal{O}_{S_0},\mathcal{O}_{S_1}} \text{ accepts}]\right] \tag{23}$$

corresponds to selecting $S_0 \sim D_s$ and $S_1 \sim D_t$. The total degree of q is obviously at most the degree of r, by the same reasoning as in the proof of Lemma 12.

To construct p, we apply the symmetrization lemma of Minsky and Papert [39] to symmetrize r, first with respect to X_0, then with respect to X_1:

$$p_0(x, X_1) = \mathbb{E}_{|S_0|=x}\, r(X_0, X_1) = \mathbb{E}_{|S_0|=x}\left[\Pr[Q^{\mathcal{O}_{S_0},\mathcal{O}_{S_1}} \text{ accepts}]\right] \tag{24}$$

$$p(x,y) = \mathbb{E}_{|S_1|=y}\, p_0(x, X_1) = \mathbb{E}_{\substack{|S_0|=x,\\ |S_1|=y}}\left[\Pr[Q^{\mathcal{O}_{S_0},\mathcal{O}_{S_1}} \text{ accepts}]\right] \tag{25}$$

The degree of p is at most the degree of r, due to Lemma 11. □

We remark that, as a consequence of their definitions in Lemma 13, p and q satisfy:

$$q(s,t) = \mathbb{E}\left[p(X,Y)\right], \tag{26}$$

where X and Y are drawn from N-trial binomial distributions with means s and t, respectively.

Symmetric Laurent polynomials. Finally, we state a useful fact about Laurent polynomials:

Lemma 14 (Symmetric Laurent polynomials)**.** *Let $\ell(x)$ be a real Laurent polynomial of positive and negative degree d that satisfies $\ell(x) = \ell(1/x)$. Then there exists a (ordinary) real polynomial q of degree d such that $\ell(x) = q(x + 1/x)$.*

Proof. $\ell(x) = \ell(1/x)$ implies that the coefficients of the x^i and x^{-i} terms are equal for all i, as otherwise $\ell(x) - \ell(1/x)$ would not equal the zero polynomial. Thus, we may write $\ell(x) = \sum_{i=0}^{d} a_i \cdot (x^i + x^{-i})$ for some coefficients a_i. So, it suffices to show that $x^i + x^{-i}$ can be expressed as a polynomial in $x + 1/x$ for all $0 \le i \le d$.

We prove by induction on i. The case $i = 0$ corresponds to constant polynomials. For $i > 0$, by the binomial theorem, observe that $(x + 1/x)^i = x^i + x^{-i} + r(x)$ where r is a degree $i - 1$ real Laurent polynomial satisfying $r(x) = r(1/x)$. By the induction assumption, r can be expressed as a polynomial in $x + 1/x$, so we have $x^i + x^{-i} = (x + 1/x)^i - r(x)$ is expressed as a polynomial in $x + 1/x$. □

4 Approximate counting with quantum samples and reflections

4.1 The Laurent polynomial method

By using Minsky–Papert symmetrization (Lemma 11), we now prove the key fact that relates quantum algorithms, of the type we're considering, to real Laurent polynomials in one variable. The following lemma generalizes the connection between quantum algorithms and real polynomials established by Beals et al. [8].

Lemma 21. *Let Q be a quantum algorithm that makes T queries to $\mathcal{O}_S$, uses R_1 copies of $|S\rangle$, and makes R_2 uses of the unitary $\mathcal{R}_S$. Let $R := R_1 + 2R_2$. For $k \in \{1, \ldots, N\}$, let*

$$q(k) := \mathbb{E}_{|S|=k}\left[\Pr\left[Q^{\mathcal{O}_S,\mathcal{R}_S}\left(|S\rangle^{\otimes R_1}\right) \text{ accepts}\right]\right]. \tag{27}$$

Then q can be written a univariate Laurent polynomial, with maximum exponent at most $2T + R$ and minimum exponent at least $-R$.

Proof. Let $|\psi_{\text{initial}}\rangle$ denote the initial state of the algorithm, which we can write as

$$|\psi_{\text{initial}}\rangle = |S\rangle^{\otimes R_1} = \left(\frac{1}{\sqrt{|S|}}\sum_{i\in S}|i\rangle\right)^{\otimes R_1} = \left(\frac{1}{\sqrt{|S|}}\sum_{i\in[N]}x_i|i\rangle\right)^{\otimes R_1}$$
$$= \frac{1}{|S|^{R_1/2}}\sum_{i_1,\ldots,i_{R_1}\in[N]} x_{i_1}\cdots x_{i_{R_1}}\,|i_1,\ldots,i_{R_1}\rangle.$$

Thus, each amplitude is a complex multilinear polynomial in $X = (x_1, \ldots, x_N)$ of degree R_1, divided by $|S|^{R_1/2}$.

Throughout the algorithm, each amplitude will remain a complex multilinear polynomial in X divided by some power of $|S|$. Since $x_i^2 = x_i$ for all i, we can always maintain multilinearity without loss of generality.

Like Beals et al. [8], we now consider how the polynomial degree of each amplitude and the power of $|S|$ in the denominator change as the algorithm progresses. We have to handle 3 different kinds of unitaries that the quantum circuit may use: the membership query oracle $\mathcal{O}_S$, unitaries independent of the input, and the reflection unitary $\mathcal{R}_S$.

The first two cases are handled as in Beals et al. Since $\mathcal{O}_S$ is a unitary whose entries are degree-1 polynomials in X, each use of this unitary increases a particular amplitude's degree as a polynomial by 1 and does not change the power of $|S|$ in the denominator. Second, input-independent unitary transformations only take linear combinations of existing polynomials and hence do not increase the degree of the amplitudes or the power of $|S|$ in the denominator. Finally, we consider the reflection unitary $\mathcal{R}_S = \mathbf{1} - 2|S\rangle\langle S|$. The $(i,j)^{\text{th}}$ entry of this operator is $\delta_{ij} - \frac{2x_ix_j}{|S|} = \frac{\delta_{ij}|S| - 2x_ix_j}{|S|}$, where δ_{ij} is the Kronecker delta function. Since $|S| = \sum_i x_i$, this is a degree-2 polynomial divided by $|S|$. Hence applying this unitary will increase the degree of the amplitudes by 2 and increase the power of $|S|$ in the denominator by 1.

In conclusion, we start with each amplitude being a polynomial of degree R_1 divided by $|S|^{R_1/2}$. T queries to the membership oracle will increase the degree of each amplitude by at most T and leave the power of $|S|$ in the denominator unchanged. R_2 uses of the reflection unitary will increase the degree by at most $2R_2$ and the power of $|S|$ in the denominator by R_2. It follows that Q's final state has the form

$$|\psi_{\text{final}}\rangle = \sum_z \alpha_z(X)\,|z\rangle\,, \tag{28}$$

where each $\alpha_z(X)$ is a complex multilinear polynomial in X of degree at most $R_1 + 2R_2 + T = R + T$, divided by $|S|^{R_1/2+R_2} = |S|^{R/2}$. Since X itself is real-valued, it follows that the real and imaginary parts of $\alpha_z(X)$, considered individually, are real multilinear polynomials in X of degree at most $R+T$ divided by $|S|^{R/2}$.

Hence, if we let

$$p(X) := \Pr\left[Q^{\mathcal{O}_S,\mathcal{R}_S}\left(|S\rangle^{\otimes R_1}\right) \text{ accepts}\right], \tag{29}$$

then

$$p(X) = \sum_{\text{accepting } z} |\alpha_z(X)|^2 = \sum_{\text{accepting } z} \left(\text{Re}^2\,\alpha_z(X) + \text{Im}^2\,\alpha_z(X)\right) \tag{30}$$

is a real multilinear polynomial in X of degree at most $2(R+T)$, divided through (in every monomial) by $|S|^R = |X|^R$.

Now consider

$$q(k) := \mathbb{E}_{|X|=k}\left[p(X)\right]. \tag{31}$$

By Lemma 11, this is a real univariate polynomial in $|X|$ of degree at most $2(R+T)$, divided through (in every monomial) by $|S|^R = |X|^R$. Or said another way, it's a real Laurent polynomial in $|X|$, with maximum exponent at most $R+2T$ and minimum exponent at least $-R$. □

4.2 Upper bounds

Before proving our lower bounds on the degree of Laurent polynomials approximating $\mathsf{ApxCount}_{N,w}$, we establish some simpler *upper bounds*. We show upper bounds on Laurent polynomial degree and in the queries, samples, and reflections model.

Laurent polynomial degree of approximate counting. We now describe a *purely negative* degree Laurent polynomial of degree $O(w^{1/3})$ for approximate counting. This upper bound will serve as an important source of intuition when we prove the (matching) lower bound of Theorem 4 (see Section 4.4.3). We are thankful to user "fedja" on MathOverflow for describing this construction. [1]

Lemma 22 (fedja)**.** *For all w, there is a real polynomial p of degree $O\left(w^{1/3}\right)$ such that:*

1. *$0 \le p(1/k) \le \frac{1}{3}$ for all $k \in [w]$.*

[1] See `https://mathoverflow.net/questions/302113/real-polynomial-bounded-at-inverse-integer-points`

2. $\frac{2}{3} \leq p(1/k) \leq 1$ *for all integers* $k \geq 2w$.

3. $0 \leq p(1/k) \leq 1$ *for all* $k \in \{w+1, w+2, \ldots, 2w-1\}$.

Proof. Assuming for simplicity that w is a perfect cube, consider

$$u(x) := (1-x)(1-2x)\cdots\left(1-w^{1/3}x\right). \tag{32}$$

Notice that $\deg(u) = w^{1/3}$ and $u\left(\frac{1}{k}\right) = 0$ for all $k \in \left[w^{1/3}\right]$. Furthermore, we have $u(x) \in [0,1]$ for all $x \in \left[0, \frac{1}{w^{1/3}}\right]$, and also $u(x) \in \left[1 - O\left(\frac{1}{w^{1/3}}\right), 1\right]$ for all $x \in \left[0, \frac{1}{w}\right]$. Now, let v be the Chebyshev polynomial of degree $w^{1/3}$, affinely adjusted so that $v(x) \in [0,1]$ for all $x \in \left[0, \frac{1}{w^{1/3}}\right]$ (rather than in $[-1,1]$ for all all $|x| \leq 1$), and with a large jump between $\frac{1}{2w}$ and $\frac{1}{w}$. Then the product, $p(x) := u(x)\, v(x)$, has degree $2w^{1/3}$ and satisfies all the requirements, except possibly that the constants $\frac{1}{3}$ and $\frac{2}{3}$ in the first two requirements may be off. Composing with a constant degree polynomial corrects this, and gives a polynomial of degree $O(w^{1/3})$ that satisfies all three requirements. □

Interestingly, if we restrict our attention to purely negative degree Laurent polynomials, then a matching lower bound is not too hard to show. In the same MathOverflow post, user fedja also proves the following, which can also be shown using earlier work of Zhandry [57, Proof of Theorem 7.3]):

Lemma 23. *Let* p *be a real polynomial, and suppose that* $|p(1/k)| \leq 1$ *for all* $k \in [2w]$*, and that* $p\left(\frac{1}{w}\right) \leq \frac{1}{3}$ *while* $p\left(\frac{1}{2w}\right) \geq \frac{2}{3}$*. Then* $\deg(p) = \Omega\left(w^{1/3}\right)$.

Section 4.3 and Section 4.4 below take the considerable step of extending Lemma 23 from purely negative degree Laurent polynomials to general Laurent polynomials.

Upper bounds in the queries, samples, and reflections model. Although we showed that there is a purely negative degree Laurent polynomial of degree $O(w^{1/3})$ for $\mathsf{ApxCount}_{N,w}$, this does not imply the existence of a quantum algorithm in the queries, samples, and reflections model with similar complexity.

We now show that our lower bounds in the queries, samples, and reflections model (in Theorem 4) are tight (up to constants). This is Theorem 5 in the introduction, restated here for convenience:

Theorem 5. *There is a quantum algorithm that solves* $\mathsf{ApxCount}_{N,w}$ *with high probability using* R *copies of* $|S\rangle$ *and reflections about* $|S\rangle$*, where* $R = O\left(\min\left\{w^{1/3}, \sqrt{\frac{N}{w}}\right\}\right)$.

Proof. We describe two quantum algorithms for this problem with the two stated complexities.

The first algorithm uses $O(w^{1/3})$ samples and reflections. This algorithm is reminiscent of the original collision finding algorithm of Brassard, Høyer, and Tapp [15]. We first use $O(w^{1/3})$ copies of $|S\rangle$ to learn a set $M \subset S$ of size $w^{1/3}$ by simply measuring copies of $|S\rangle$ in the computational basis. Now we know that the ratio $|S|/|M|$ is either $w^{2/3}$ or $2w^{2/3}$. Now consider running Grover's algorithm on the set S where the elements in M are considered the "marked" elements. Grover's algorithm alternates reflections about the uniform

superposition over the set being searched, S, with an operator that reflects about the marked elements in M. The first reflection is simply $\mathcal{R}_S$, which we have access to. The second unitary can be constructed since we have an explicit description of the set M. Now Grover's algorithm can be used to distinguish whether the fraction of marked elements is $1/w^{2/3}$ or half of that, and the cost will be $O(w^{1/3})$.

The second algorithm uses $O(\sqrt{N/w})$ reflections only and no copies of $|S\rangle$. Consider running the standard approximate counting algorithm [13] that uses membership queries to S and distinguishes $|S| \leq w$ from $|S| \geq 2w$ using $O(\sqrt{N/w})$ membership queries. Observe that this algorithm starts with the state $|\psi\rangle = \frac{1}{\sqrt{N}}(|1\rangle + \cdots + |N\rangle)$, which is in $\text{span}\{|S\rangle, |\bar{S}\rangle\}$, and only uses reflections about $|\psi\rangle$ and membership queries to $|S\rangle$ in the form of a unitary that maps $|i\rangle$ to $-|i\rangle$ when $i \in S$. This means the state of the algorithm remains in $\text{span}\{|S\rangle, |\bar{S}\rangle\}$ at all times. Within this subspace, a membership query to S is the same as a reflection about $|S\rangle$. Hence we can replace membership queries with the reflection operator to get an approximate counting algorithm that only uses $O(\sqrt{N/w})$ reflections and no copies of $|S\rangle$. □

Note that both the algorithms presented above generalize to the situation where we want to distinguish $|S| = w$ from $|S| = (1+\varepsilon)w$. For the first algorithm, we now pick a subset M of size $w^{1/3}/\varepsilon^{2/3}$. Now we want to $(1+\varepsilon)$-approximate the fraction of marked elements, which is either $1/(w\varepsilon)^{2/3}$ or $(1+\varepsilon)^{-1}$ times that. This can be done with approximate counting [13, Theorem 15], and the cost will be $O\left(\frac{1}{\varepsilon}(w\varepsilon)^{1/3}\right) = O\left(\frac{w^{1/3}}{\varepsilon^{2/3}}\right)$. The second algorithm is simpler to generalize, since we simply plug in the query complexity of ε-approximate counting, which is $O\left(\frac{1}{\varepsilon}\sqrt{\frac{N}{w}}\right)$.

4.3 Lower bound using the explosion argument

We now show a weaker version of Theorem 4 using the explosion argument described in the introduction. The difference between the following theorem and Theorem 4 is the exponent of w in the lower bound.

Theorem 24. *Let Q be a quantum algorithm that makes T queries to the membership oracle for S, and uses a total of R copies of $|S\rangle$ and reflections about $|S\rangle$. If Q decides whether $|S| = w$ or $|S| = 2w$ with success probability at least $2/3$, promised that one of those is the case, then either*

$$T = \Omega\left(\sqrt{\frac{N}{w}}\right) \qquad \text{or} \qquad R = \Omega\left(\min\left\{w^{1/4}, \sqrt{\frac{N}{w}}\right\}\right). \tag{33}$$

Proof. Since we neglect multiplicative constants in our lower bounds, let us allow the algorithm to use up to R copies of $|S\rangle$ and R uses of $\mathcal{R}_S$. Let

$$q(k) := \mathbb{E}_{|S|=k}\left[\Pr\left[Q^{\mathcal{O}_S, \mathcal{R}_S}\left(|S\rangle^{\otimes R}\right) \text{ accepts}\right]\right]. \tag{34}$$

Then by Lemma 21, we can write q as a Laurent polynomial:

$$q(k) = u(k) + v(1/k), \tag{35}$$

where u is a real polynomial in k with $\deg(u) = O(T+R)$, and v is a real polynomial in $1/k$ with $\deg(v) = O(R)$. So to prove the theorem, it suffices to show that either $\deg(u) = \Omega\left(\sqrt{\frac{N}{w}}\right)$, or else $\deg(v) = \Omega\left(w^{1/4}\right)$. To do so, we'll assume that $\deg(u) = o\left(\sqrt{\frac{N}{w}}\right)$ and $\deg(v) = o\left(w^{1/4}\right)$, and derive a contradiction.

Our high-level strategy is as follows: we'll observe that, if approximate counting is being successfully solved, then either u or v must attain a large first derivative somewhere in its domain. By the approximation theory lemmas that we proved in Section 2.1, this will force that polynomial to have a large range—even on a subset of integer (or inverse-integer) points. But the sum, $u(k) + v(1/k)$, is bounded in $[0,1]$ for all $k \in [N]$. So if one polynomial has a large range, then the other does too. But this forces the *other* polynomial to have a large derivative somewhere in its domain, and therefore (by approximation theory) to have an even larger range, forcing the first polynomial to have an even larger range to compensate, and so on. As long as $\deg(u)$ and $\deg(v)$ are both small enough, this endless switching will force both u and v to attain *unboundedly* large values—with the fact that one polynomial is in k, and the other is in $1/k$, crucial to achieving the desired "explosion." Since u and v are polynomials on compact sets, such unbounded growth is an obvious absurdity, and this will give us the desired contradiction.

In more detail, we will study the following quantities.

$$\begin{array}{ll}
G_u := \max_{x,y \in \left[\sqrt{w}, 2w\right]} |u(x) - u(y)| & G_v := \max_{x,y \in \left[\frac{1}{N}, \frac{1}{w}\right]} |v(x) - v(y)| \\
\Delta_u := \max_{x \in \left[\sqrt{w}, 2w\right]} |u'(x)| & \Delta_v := \max_{x \in \left[\frac{1}{N}, \frac{1}{w}\right]} |v'(x)| \\
H_u := \max_{x,y \in \left[\sqrt{w}, N\right]} |u(x) - u(y)| & H_v := \max_{x,y \in \left[\frac{1}{N}, \frac{1}{\sqrt{w}}\right]} |v(x) - v(y)| \\
I_u := \max_{x,y \in [w,N]} |u(x) - u(y)| & I_v := \max_{x,y \in \left[\frac{1}{2w}, \frac{1}{\sqrt{w}}\right]} |v(x) - v(y)| \\
L_u := \max_{x,y \in \{w,\ldots,N\}} |u(x) - u(y)| & L_v := \max_{x,y \in \left\{\sqrt{w},\ldots,2w\right\}} \left|v\left(\frac{1}{x}\right) - v\left(\frac{1}{y}\right)\right|
\end{array} \tag{36}$$

We have $0 \leq q(k) \leq 1$ for all $k \in [N]$, since in those cases $q(k)$ represents a probability. Since Q solves approximate counting, we also have $q(w) \leq \frac{1}{3}$ and $q(2w) \geq \frac{2}{3}$. This means in particular that either

(i) $u(2w) - u(w) \geq \frac{1}{6}$, and hence $G_u \geq \frac{1}{6}$, or else

(ii) $v\left(\frac{1}{2w}\right) - v\left(\frac{1}{w}\right) \geq \frac{1}{6}$, and hence $G_v \geq \frac{1}{6}$.

We will show that either case leads to a contradiction.

We have the following inequalities regarding u:

$$\begin{array}{ll}
G_u \geq L_v - 1 & \text{by the boundedness of } q \\
\Delta_u \geq \frac{G_u}{2w} & \text{by basic calculus} \\
H_u \geq \frac{\Delta_u\left(N - \sqrt{w}\right)}{\deg(u)^2} & \text{by Lemma 6} \\
I_u \geq \frac{H_u}{2} & \text{by Corollary 8} \\
L_u \geq \frac{I_u}{2} & \text{by Lemma 9}
\end{array} \tag{37}$$

Here the fourth inequality uses the fact that, setting $\varepsilon := \frac{\sqrt{w}}{N}$, we have $\deg(u) = o\left(\frac{1}{\sqrt{\varepsilon}}\right)$ (thereby satisfying the hypothesis of Corollary 8), while the fifth inequality uses the fact that $\deg(u) = o\left(\sqrt{N}\right)$.

Meanwhile, we have the following inequalities regarding v:

$$\begin{array}{ll} G_v \geq L_u - 1 & \text{by the boundedness of } q \\ \Delta_v \geq G_v w & \text{by basic calculus} \\ H_v \geq \frac{\Delta_v \left(\frac{1}{\sqrt{w}} - \frac{1}{N}\right)}{\deg(v)^2} & \text{by Lemma 6} \\ I_v \geq \frac{H_v}{2} & \text{by Corollary 8} \\ L_v \geq \frac{I_v}{2} & \text{by Lemma 9} \end{array} \tag{38}$$

Here the fourth inequality uses the fact that, setting $\varepsilon := \frac{1/2w}{1/\sqrt{w}} = \frac{1}{2\sqrt{w}}$, we have $\deg(v) = o\left(\frac{1}{\sqrt{\varepsilon}}\right)$ (thereby satisfying the hypothesis of Corollary 8). The fifth inequality uses the fact that, if we set $V(x) := v(x/w)$, then the situation satisfies the hypothesis of Lemma 9: we are interested in the range of V on the interval $\left[\frac{1}{2}, \sqrt{w}\right]$, compared to its range on discrete points $\frac{w}{\sqrt{w}}, \frac{w}{\sqrt{w}+1}, \ldots, \frac{w}{2w}$ that are spaced at most 1 apart from each other; and we also have $\deg(V) = \deg(v) = o\left(w^{1/4}\right)$.

All that remains is to show that, if we insert either $G_u \geq \frac{1}{6}$ or $G_v \geq \frac{1}{6}$ into the coupled system of inequalities above, then we get unbounded growth and the inequalities have no solution. Let us collapse the two sets of inequalities to

$$L_u \geq \frac{1}{4} \frac{N - \sqrt{w}}{\deg(u)^2} \frac{G_u}{2w} = \Omega\left(\frac{N}{w \deg(u)^2} G_u\right),$$
$$L_v \geq \frac{1}{4} \frac{\frac{1}{\sqrt{w}} - \frac{1}{N}}{\deg(v)^2} G_v w = \Omega\left(\frac{\sqrt{w}}{\deg(v)^2} G_v\right).$$

Hence

$$G_u \geq L_v - 1 = \Omega\left(\frac{\sqrt{w}}{\deg(v)^2} G_v\right) - 1,$$
$$G_v \geq L_u - 1 = \Omega\left(\frac{N}{w \deg(u)^2} G_u\right) - 1.$$

By the assumption that $\deg(v) = o\left(w^{1/4}\right)$ and $\deg(u) = o\left(\sqrt{\frac{N}{w}}\right)$, we have $\frac{\sqrt{w}}{\deg(v)^2} \gg 1$ and $\frac{N}{w \deg(u)^2} \gg 1$. Plugging in $G_u \geq \frac{1}{6}$ or $G_v \geq \frac{1}{6}$, this is enough to give us unbounded growth. □

4.4 Lower bound using dual polynomials

In this section we use the method of dual polynomials to establish our main result, Theorem 4, restated for convenience:

Theorem 4. *Let Q be a quantum algorithm that makes T queries to the membership oracle for S, and uses a total of R copies of $|S\rangle$ and reflections about $|S\rangle$. If Q decides whether $|S| = w$ or $|S| = 2w$ with high probability, promised that one of those is the case, then either*

$$T = \Omega\left(\sqrt{\frac{N}{w}}\right) \qquad \text{or} \qquad R = \Omega\left(\min\left\{w^{1/3}, \sqrt{\frac{N}{w}}\right\}\right). \tag{1}$$

Let $p(r)$ be a univariate Laurent polynomial of negative degree D_1 and positive degree D_2. That is, let $p(r)$ be of the form

$$p(r) = a_0/r^{D_1} + a_1/r^{D_1-1} + \cdots + a_{D_1-1}/r + a_{D_1} + a_{D_1+1} \cdot r + \cdots + a_{D_2+D_1} \cdot r^{D_2}. \tag{39}$$

Theorem 4 follows by combining the Laurent polynomial method (Lemma 21) and the following theorem.

Theorem 25. *Let $\varepsilon < 1$. Suppose that p has negative degree D_1 and positive degree D_2 and satisfies the following properties.*

- $|p(w) - 1| \leq \varepsilon$
- $|p(2w) + 1| \leq \varepsilon$
- $|p(\ell)| \leq 1 + \varepsilon$ *for all* $\ell \in \{1, 2, \ldots, n\}$

Then either $D_1 \geq \Omega\left(w^{1/3}\right)$ *or* $D_2 \geq \Omega\left(\sqrt{N/w}\right)$.

In fact, our proof of Theorem 25 will show that the lower bound holds even if $|p(\ell)| \leq 1 + \varepsilon$ only for $\ell \in \{w^{1/3}, w^{1/3} + 1, \ldots, w\} \cup \{2w, 2w + 1, \ldots, N\}$. We refer to a Laurent polynomial p satisfying the three properties of Theorem 25 as an *approximation for approximate counting*.

Proof of Theorem 25. Let p be any Laurent polynomial satisfying the hypothesis of Theorem 25 . We begin by transforming p into a (standard) polynomial q in a straightforward manner. This transformation is captured in the following lemma, whose proof is so simple that we omit it.

Lemma 26. *If p satisfies the properties of Theorem 25, then the polynomial $q(r) = p(r) \cdot r^{D_1} = a_0 + a_1 r + \cdots + a_{D_1+D_2} r^{D_1+D_2}$ is a (standard) polynomial of degree at most $D_1 + D_2$, and q satisfies the following three properties.*

- $\left|q(w) - w^{D_1}\right| \leq \varepsilon \cdot w^{D_1}$
- $\left|q(2w) + (2w)^{D_1}\right| \leq \varepsilon \cdot (2w)^{D_1}$
- $|q(\ell)| \leq (1 + \varepsilon)\, \ell^{D_1}$ *for all* $\ell \in \{1, 2, \ldots, N\}$

We now turn to showing that, for any constant $\varepsilon < 1$, no polynomial q can satisfy the conditions of Lemma 26 unless $D_1 \geq \Omega(w^{1/3})$ or $D_2 \geq \Omega\left(\sqrt{N/w}\right)$.

Consider the following linear program. The variables of the linear program are ε, and the $D_2 + D_1 + 1$ coefficients of q.

$$\begin{array}{ll} \text{minimize} & \varepsilon \\ \text{such that} & \\ & |q(w) - w^{D_1}| \leq \varepsilon \cdot w^{D_1} \\ & |q(2w) + (2w)^{D_1}| \leq \varepsilon \cdot (2w)^{D_1} \\ & |q(\ell)| \leq (1 + \varepsilon) \cdot \ell^{D_1} \text{ for all } \ell \in \{1, 2, \ldots, N\} \\ & \varepsilon \geq 0 \end{array} \tag{40}$$

Standard manipulations reveal the dual.

$$\boxed{\begin{array}{l}\text{maximize } \phi(w)\cdot w^{D_1} - \phi(2w)\cdot(2w)^{D_1} - \sum_{\ell\in\{1,\ldots,N\},\ell\notin\{w,2w\}} |\phi(\ell)|\cdot\ell^{D_1} \\ \text{such that} \\ \sum_{\ell=1}^{N}\phi(\ell)\cdot\ell^{j} = 0 \text{ for } j = 0,1,2,\ldots,D_1+D_2 \\ \sum_{\ell=1}^{N}|\phi(\ell)|\cdot\ell^{D_1} = 1 \\ \phi\colon \mathbb{R}\to\mathbb{R}\end{array}} \tag{41}$$

Theorem 25 will follow if we can exhibit a solution ϕ to the dual linear program achieving value $\varepsilon > 0$, for some setting of $D_1 \geq \Omega(w^{1/3})$ and $D_2 \geq \Omega\left(\sqrt{N/w}\right)$.[2] We now turn to this task.

4.4.1 Constructing the dual solution

For a set $T \subseteq \{0,1,\ldots,N\}$, define

$$Q_T(t) = \prod_{i=0,1,\ldots,N, i\notin T} (t-i). \tag{42}$$

Let $c > 2$ be an integer constant that we will choose later (the bigger we choose c to be, the better the objective value achieved by our final dual witness. But choosing a bigger c will also lower the degrees D_1, D_2 of Laurent polynomials against which our lower bound will hold).

We now define two sets T_1 and T_2. The size of T_1 will be

$$d_1 := \lfloor (w/c)^{1/3} \rfloor = \Theta\left(w^{1/3}\right) \tag{43}$$

and the size of T_2 will be d_2 for

$$d_2 := \lfloor \sqrt{N/(cw)} \rfloor = \Theta\left(\sqrt{N/w}\right). \tag{44}$$

Let

$$T_1 = \left\{ \lfloor w/(ci^2) \rfloor \colon i = 1,2,\ldots,d_1 \right\} \tag{45}$$

and

$$T_2 = \left\{ c\cdot i^2\cdot w \colon i = 1,2,\ldots,d_2 := \sqrt{N/(cw)} \right\}. \tag{46}$$

Finally, define

$$T = \{w, 2w\} \cup T_1 \cup T_2. \tag{47}$$

At last, define $\Phi\colon \{0,1,\ldots,N\}\to\mathbb{R}$ via

$$\Phi(t) = (-1)^t \cdot \binom{N}{t} \cdot Q_T(t). \tag{48}$$

[2] We will alternatively refer to such dual solutions ϕ as *dual witnesses*, since they act as a witness to the fact that any low-degree Laurent polynomial p approximating the approximate counting problem must have large error.

Our final dual solution ϕ will be a scaled version of Φ. Specifically, Φ itself does not satisfy the second constraint of the dual linear program, that $\sum_{\ell=1}^{N} |\Phi(\ell)| \cdot \ell^{D_1} = 1$. So letting

$$C = \sum_{\ell=1}^{N} |\Phi(\ell)| \cdot \ell^{D_1}, \tag{49}$$

our final dual witness ϕ will be Φ/C.

The sizes of T_1 and T_2. Clearly, under the above definition of T_2, $|T_2| = d_2$ as claimed above. It is not as immediately evident that $|T_1| = d_1$: to establish this, we must show that for distinct $i, j \in \{1, 2, \ldots, d_1\}$, $\lfloor w/(ci^2) \rfloor \neq \lfloor w/(cj^2) \rfloor$. This is handled in the following easy lemma.

Lemma 27. *Let $i \neq j$ be distinct numbers in $\{1, \ldots, d_1\}$ and $c > 2$ be a constant. Then as long as $d_1 < (w/c)^{1/3}$, it holds that $\lfloor w/(ci^2) \rfloor \neq \lfloor w/(cj^2) \rfloor$.*

Proof. Assume without loss of generality that $i > j$. Then $w/(cj^2) - w/(ci^2)$ is clearly minimized when $i = d_1$ and $j = i - 1$. For the remainder of the proof, fix $i = d_1$. In this case,

$$\begin{aligned} w/(cj^2) - w/(ci^2) \geq w/\left(c(i-1)^2\right) - w/\left(ci^2\right) &= \frac{wi^2 - w\left(i-1\right)^2}{c \cdot i^2 \cdot \left(i-1\right)^2} \\ &= \frac{w}{c} \cdot \frac{2i-1}{i^2\left(i-1\right)^2} \geq \frac{w}{c} \cdot \frac{2i-1}{i^4} \geq \frac{w}{ci^3} \geq 1. \end{aligned} \tag{50}$$

Here, the final inequality holds because $i^3 = d_1^3 \leq w/c$.

Equation (50) implies the lemma, as two numbers whose difference is at least 1 cannot have the same integer floor. □

Lemma 27 is false for $d_1 = \omega(w^{1/3})$, highlighting on a technical level why one cannot choose d_1 larger than $\Theta(w^{1/3})$ without the entire construction and analysis of Φ breaking down.

4.4.2 Intuition: "gluing together" two simpler dual solutions

Before analyzing the dual witnesses Φ and ϕ constructed in Equation (48) and Equation (49), in this subsection and the next, we provide detailed intuition for why the definitions of Φ and ϕ are natural, and briefly overview their analysis.

A dual witness for purely positive degree (i.e., approximate degree). Suppose we were merely interested in showing an approximate degree lower bound of $\Omega(\sqrt{N/w})$ for approximate counting (i.e., a lower bound on the degree of traditional polynomials that distinguish input w from $2w$, and are bounded at all other integer inputs in $1, \ldots, N$). This is equivalent to exhibiting a solution to the dual linear program with $D_1 = 0$. A valid dual witness ϕ_1 for this simpler case is to also use Equation (48), but to set

$$T = \{w, 2w\} \cup T_2, \tag{51}$$

rather than $T = \{w, 2w\} \cup T_1 \cup T_2$.

We will explain intuition for why Equation (51) is a valid dual solution for the approximate degree of approximate counting in the next subsection. For now, we wish to explain how this construction relates to prior work. In [18], for any constant $\delta > 0$, a dual witness is given for the fact that the $(1-\delta)$-approximate degree of OR is $\Omega(\sqrt{N})$. This dual witness *nearly* corresponds to the above, with $w = 1$. Specifically, Bun and Thaler [18] use the set $T = \{0,1\} \cup \{ci^2 \colon i = 1, 2, \ldots, \sqrt{N/c}\}$, and they show that almost all of the "mass" of this dual witness is located on the inputs 0 and 1, i.e.,

$$|\Phi(0)| + |\Phi(1)| \geq (1-\delta) \cdot \sum_{i=2}^{N} |\Phi(i)| \,. \tag{52}$$

Here, the bigger c is chosen to be, the smaller the value of δ for which Equation (52) holds.

In the case of $w = 1$, our dual witness for approximate counting differs from this only in that $\{0,1\}$ is replaced with $\{1,2\}$. This is because, in order to show a lower bound for distinguishing input $w = 1$ from input $2w = 2$, we want almost all of the mass to be on inputs $\{1,2\}$ rather than $\{0,1\}$ (this is what will ensure that the objective function of the dual linear program is large).

For general w, we want most of the mass of ψ to be concentrated on inputs w and $2w$. Accordingly, relative to the $w = 1$ case, we effectively multiply *all* points in T by w, and one can show that this does not affect the calculation regarding concentration of mass.

A dual witness for purely negative degree. Now, suppose we were merely interested in showing that Laurent polynomials of *purely negative* degree require degree $\Omega(w^{1/3})$ to approximate the approximate counting problem. This is equivalent to exhibiting a solution to the dual linear program with $D_2 = 0$. Then a valid dual witness ϕ_2 for this simpler case is to also use Equation (48), but to set

$$T = \{w, 2w\} \cup T_1. \tag{53}$$

Again, we will give intuition for why this is a valid dual solution in the next subsection (Section 4.4.3). For now, we wish to explain how this construction relates to prior work. Essentially, the $\Omega\left(w^{1/3}\right)$-degree lower bound for Laurent polynomials with *only negative* powers was proved by Zhandry [57, Theorem 7.3]. Translating Zhandry's theorem into our setting is not entirely trivial, and he did not explicitly construct a solution to our dual linear program. However (albeit with significant effort), one can translate his argument to our setting to show that Equation (53) gives a valid dual solution to prove a lower bound against Laurent polynomials with only negative powers.

Gluing them together. The above discussion explains that the key ideas for constructing dual solutions ϕ_1, ϕ_2 witnessing degree lower bounds for Laurent polynomials of *only negative* or *only positive* powers were essentially already known, or at least can be extracted from prior work with enough effort. In this work, we are interested in proving lower bounds for Laurent polynomials with both positive and negative powers. Our dual solution Φ essentially just "glues together" the dual solutions that can be derived from prior work. By

this, we mean that the set T of integer points on which our Φ is nonzero is the *union* of the corresponding sets for ϕ_1 and ϕ_2 individually. Moreover, this union is nearly disjoint, as the only points in the intersection of the two sets being unioned are w and $2w$.

Overview of the analysis. To show that we have constructed a valid solution to the dual linear program (Equation (41)), we must establish that (a) Φ is uncorrelated with every polynomial of degree at most $D_1 + D_2$ and (b) Φ is well-correlated with any function g that evaluates to $+1$ on input w, to -1 on input $2w$, and is bounded in $[-1, 1]$ elsewhere. In (b), the correlation is taken with respect to an appropriate weighting of the inputs, that on input $\ell \in [N]$ places mass proportional to ℓ^{D_1}.

The definition of Φ as a "gluing together" of ϕ_1 and ϕ_2 turns out, in a straightforward manner, to ensure that Φ is uncorrelated with polynomials of degree at $D_1 + D_2$. All that remains is to show that Φ is well-correlated with g under the appropriate weighting of inputs. This turns out to be technically demanding, but ultimately can be understood as stemming from the fact that ϕ_1 and ϕ_2 are individually well-correlated with g (albeit, in the case of ϕ_2, under a *different* weighting of the inputs than the weighting that is relevant for Φ).

4.4.3 Intuition via complementary slackness

We now attempt to lend some insight into why the dual witnesses ϕ_1 and ϕ_2 for the purely positive degree and purely negative degree take the form that they do. This section is deliberately slightly imprecise in places, and builds on intuition that has been put forth in prior works proving approximate degree lower bounds via dual witnesses [18, 55, 17].

Notice that ϕ_1 is precisely defined so that $\phi_1(i) = 0$ for any $i \notin \{w, 2w\} \cup T_2$, and similarly $\phi_2(i) = 0$ for any $i \notin \{w, 2w\} \cup T_1$. The intuition for why this is reasonable comes from complementary slackness, which states that an optimal dual witness should equal 0 except on inputs that correspond to primal constraints that are *made tight by an optimal primal solution*. By "constraints made tight by an optimal primal solution", we mean constraints that, for the optimal primal solution, hold with equality rather than (strict) inequality.

Unpacking that statement, this means the following. Suppose that q is an optimal solution to the primal linear program of Section 4.4, meaning it minimizes the error ε amongst all polynomials of the same same degree. The constraints made tight by q are precisely those inputs ℓ at which q hits its "maximum error" (e.g., an input ℓ such that $|q(\ell)| = (1 + \varepsilon) \cdot \ell^{D_1}$). We call these inputs *maximum-error* inputs for q. Complementary slackness says that there is an optimal solution to the dual linear program (Equation (41)) that equals 0 at all inputs that are not maximum-error inputs for q.

In both the purely positive degree case, and the purely negative degree case, we know roughly what primal optimal solutions q look like, and moreover we know what roughly their maximum-error points look like. In the first case, the maximum-error points are well-approximated by the points in T_2, and in the purely negative degree case, the maximum error points are well-approximated by the points in T_1. Let us explain.

Purely positive degree case. Let T_d be the degree d Chebyshev polynomial of the first kind. It can be seen that $P(\ell) = T_{\sqrt{N}}(1 + 2/N - \ell/N)$ satisfies $P(1) \geq 2$, while $|P(\ell)| \leq 1$ for $\ell = 2, 3, \ldots, N$. That is, up to scaling, P approximates the approximate counting problem for $w = 1$, and its known that its degree is within a constant factor of optimal.

It is known that the extreme points of T_d are of the following form, for $k = 1, \ldots, d$:

$$\cos\left(\frac{(2k-1)}{2d}\pi\right) \approx 1 - k^2/(2d^2), \tag{54}$$

where the approximation uses the Taylor expansion of the cosine function around 0. Equation (54) means that the extreme points of P are roughly those inputs ℓ such that $1 + 2/N - \ell/N \approx 1 - k^2/(2d^2)$, where $d = \sqrt{N}$. Such ℓ are roughly of the form $\ell \approx c \cdot i^2$ for some constant c, as i ranges from 1 up to $\Theta(N^{1/2})$.

More generally, when $w \geq 1$, an asymptotically optimal approximation for distinguishing input w from $2w$ is $P(\ell) = T_{\sqrt{N/w}}(1 + 2w/N - \ell/(wN))$. The extreme points of P are roughly of the form $\ell \approx c \cdot i^2 \cdot w$ for some constant c, as i ranges from 1 up to $\Theta(\sqrt{N/w})$, which is exactly the form of the points in our set T_2.

Purely negative degree case. In Lemma 22 , we exhibited a simple, purely negative degree Laurent polynomial p (i.e., $p(\ell)$ is a standard polynomial in $1/\ell$) with degree $D_1 = w^{1/3}$ that solves the approximate counting problem (the construction is due to MathOverflow user "fedja"). Roughly speaking, p can be written as a product $p(\ell) = u(\ell) \cdot v(\ell)$, where $u(\ell)$ has the roots $\ell = 1, 2, \ldots, w^{1/3}$, and $v(\ell)$ is (an affine transformation) of a Chebyshev polynomial of degree $w^{1/3}$, applied to $1/\ell$. One can easily look at this construction and see that $p(\ell)$ outputs *exactly* the correct value on inputs $\{1, 2, \ldots, w^{1/3}\}$, so these are not maximum error points for p. Moreover, the analysis of the maximum error points for Chebyshev polynomials above can be applied to show that the maximum error points of p are roughly of the form ℓ such that $1/\ell = c \cdot i^2/w$ for some constant c, with i ranging from 1 up to $\Theta(w^{1/3})$. This means that the extreme points are roughly of the form $\ell \approx \frac{w}{ci^2}$, which is why our set T_1 consists of points of the form $\lfloor \frac{w}{ci^2} \rfloor$ (the floors are required because we are proving lower bounds against polynomials whose behavior is only constrained at integer inputs).

4.4.4 Analysis of the dual solution Φ

Lemma 28. *Let $d_1 = |T_1|$ and $d_2 = |T_2|$. Then for any $j = 0, 1, \ldots, d_1 + d_2$, it holds that*

$$\sum_{\ell=1}^{N} \Phi(\ell) \cdot \ell^j = 0.$$

Proof. A basic combinatorial fact is that for any polynomial Q of degree at most $N - 1$, the following identity holds:

$$\sum_{\ell=0}^{N} \binom{N}{\ell} (-1)^\ell Q(\ell) = 0. \tag{55}$$

Observe that for any $j \leq d_1 + d_2 + 1$,

$$Q_T(\ell) \cdot \ell^j \text{ is a polynomial in } \ell \text{ of degree at most } N-1. \tag{56}$$

Furthermore, $\Phi(0) = 0$, because $0 \notin T$. Hence

$$\sum_{\ell=0}^{N} \binom{N}{\ell} (-1)^\ell Q_T(\ell) \cdot \ell^j = \sum_{\ell=1}^{N} \binom{N}{\ell} (-1)^\ell Q_T(\ell) \cdot \ell^j. \tag{57}$$

Thus, we can calculate:

$$\begin{aligned}\sum_{\ell=1}^{N} \Phi(\ell) \cdot \ell^j &= \sum_{\ell=1}^{N} (-1)^\ell \cdot \binom{N}{\ell} \cdot Q_T(\ell) \cdot \ell^j \\ &= \sum_{\ell=0}^{N} (-1)^\ell \cdot \binom{N}{\ell} \cdot Q_T(\ell) \cdot \ell^j = 0.\end{aligned}$$

Here, the second equality follows from Equation (57), while the third follows from Equations (55) and (56). □

Let us turn to analyzing Φ's value on various inputs. Clearly the following condition holds:

$$\Phi(\ell) = 0 \text{ for all } \ell \notin T. \tag{58}$$

Next, observe that for any $r \in T$,

$$|\Phi(r)| = N! \cdot \frac{1}{\prod_{j \in T, j \neq r} |r - j|}.$$

Consider any quantity $c \cdot i^2 \cdot w \in T_2$. Then

$$\begin{aligned}\left|\Phi(c \cdot w \cdot i^2)\right| / |\Phi(w)| &= \frac{\prod_{j \in T, j \neq w} |w - j|}{\prod_{j \in T, j \neq c \cdot i^2 \cdot w} |w \cdot c \cdot i^2 - j|} \\ &= \frac{|w - 2w| \cdot \left(\prod_{j=1}^{d_2} \left|w - c \cdot j^2 \cdot w\right|\right) \cdot \left(\prod_{j=1}^{d_1} \left(w - \left\lfloor \frac{w}{cj^2} \right\rfloor\right)\right)}{|c \cdot i^2 \cdot w - w| \cdot |c \cdot i^2 \cdot w - 2w| \cdot \left(\prod_{j=1, j \neq i}^{d_2} |w \cdot c \cdot i^2 - w \cdot c \cdot j^2|\right) \cdot \left(\prod_{j=1}^{d_1} \left(w \cdot c \cdot i^2 - \left\lfloor \frac{w}{c \cdot j^2} \right\rfloor\right)\right)} \\ &= \frac{c^{d_2} \cdot \left(\prod_{j=1}^{d_2} \left(j^2 - \frac{1}{c}\right)\right) \cdot \prod_{j=1}^{d_1} \left(w - \left\lfloor \frac{w}{c \cdot j^2} \right\rfloor\right)}{(ci^2 - 1) \cdot (ci^2 - 2) \cdot c^{d_2 - 1} \cdot \left(\prod_{j=1, j \neq i}^{d_2} |i^2 - j^2|\right) \cdot \left(\prod_{j=1}^{d_1} \left(w \cdot c \cdot i^2 - \left\lfloor \frac{w}{c \cdot j^2} \right\rfloor\right)\right)} \\ &\leq \frac{c \cdot \left(\prod_{j=1}^{d_2} \left(j^2 - \frac{1}{c}\right)\right) \cdot \prod_{j=1}^{d_1} \left(w - \left\lfloor \frac{w}{c \cdot j^2} \right\rfloor\right)}{(ci^2 - 1) \cdot (ci^2 - 2) \cdot \left(\prod_{j=1, j \neq i}^{d_2} |i^2 - j^2|\right) \cdot \left(\prod_{j=1}^{d_1} \left(w \cdot c \cdot i^2 - \frac{w}{c \cdot j^2}\right)\right)}\end{aligned} \tag{59}$$

Now, observe that

$$\prod_{j=1}^{d_1} \left(w - \left\lfloor \frac{w}{c \cdot j^2} \right\rfloor\right) \leq \prod_{j=1}^{d_1} \left(w - \frac{w}{cj^2} + 1\right) = \prod_{j=1}^{d_1} w \cdot \left(1 - \frac{1}{cj^2}\right) \cdot \left(1 + \frac{1}{w \cdot \left(1 - \frac{1}{cj^2}\right)}\right)$$

$$\leq \prod_{j=1}^{d_1} w \cdot \left(1 - \frac{1}{cj^2}\right)\left(1 + \frac{1}{(1-1/c)\cdot w}\right) \leq \left(\prod_{j=1}^{d_1} w \cdot \left(1 - \frac{1}{cj^2}\right)\right) \cdot (1 + o(1)) \,. \tag{60}$$

Hence, we see that Expression (59) is bounded by

$$\begin{aligned}
&\frac{c \cdot \left(\prod_{j=1}^{d_2}\left(j^2 - \frac{1}{c}\right)\right) \cdot \left(\prod_{j=1}^{d_1}\left(1 - \frac{1}{c\cdot j^2}\right)\right) \cdot (1 + o(1))}{(ci^2-1)\cdot(ci^2-2)\cdot\left(\prod_{j=1, j\neq i}^{d_2} |i^2 - j^2|\right)\cdot\left(\prod_{j=1}^{d_1}\left(c\cdot i^2 - \frac{1}{c\cdot j^2}\right)\right)} \\
&\leq \frac{c\cdot (d_2!)^2 \cdot \left(\prod_{j=1}^{d_1}\left(1 - \frac{1}{c\cdot j^2}\right)\right)\cdot(1+o(1))}{(ci^2-1)\cdot(ci^2-2)\cdot\left(\prod_{j=1,j\neq i}^{d_2}|i-j|\,|i+j|\right)\cdot(c\cdot i^2)^{d_1}\cdot\left(\prod_{j=1}^{d_1}\left(1-\frac{1}{c^2\cdot i^2\cdot j^2}\right)\right)} \\
&= \frac{c\cdot (d_2!)^2\cdot 2i^2\cdot\left(\prod_{j=1}^{d_1}\left(1-\frac{1}{c\cdot j^2}\right)\right)\cdot(1+o(1))}{(ci^2-1)\cdot(ci^2-2)\cdot(d_2+i)!\,(d_2-i)!\cdot(c\cdot i^2)^{d_1}\cdot\left(\prod_{j=1}^{d_1}\left(1-\frac{1}{c^2\cdot i^2\cdot j^2}\right)\right)} \\
&\leq \frac{c\cdot 2i^2\cdot(d_2!)^2\cdot(1+o(1))}{(ci^2-1)\,(ci^2-2)\cdot(d_2+i)!\,(d_2-i)!\cdot(c\cdot i^2)^{d_1}} \leq \frac{2\,(1+o(1))}{\left(1-\frac{1}{c\cdot i^2}\right)\cdot(c\cdot i^2-2)\cdot(c\cdot i^2)^{d_1}}.
\end{aligned} \tag{61}$$

In the penultimate inequality, we used the fact that $\frac{(d_2!)^2}{(d_2+i)!(d_2-i)!} = \frac{\binom{2d_2}{d_2+i}}{\binom{2d_2}{d_2}} \leq 1$.

Next, consider any quantity $\left\lfloor \frac{w}{c\cdot i^2} \right\rfloor \in T_1$. Then

$$\begin{aligned}
&\left|\Phi\left(\left\lfloor\frac{w}{c\cdot i^2}\right\rfloor\right)\right| / \left|\Phi(w)\right| \\
&= \frac{|w-2w|\left(\prod_{j=1}^{d_2}|w-cj^2w|\right)\left(\prod_{j=1}^{d_1}\left(w-\left\lfloor\frac{w}{cj^2}\right\rfloor\right)\right)}{\left(w-\left\lfloor\frac{w}{c\cdot i^2}\right\rfloor\right)\cdot\left(2w-\left\lfloor\frac{w}{c\cdot i^2}\right\rfloor\right)\left(\prod_{j=1}^{d_2}\left(w\cdot c\cdot j^2-\left\lfloor\frac{w}{c\cdot i^2}\right\rfloor\right)\right)\prod_{j=1,j\neq i}^{d_1}\left|\left\lfloor\frac{w}{c\cdot i^2}\right\rfloor-\left\lfloor\frac{w}{c\cdot j^2}\right\rfloor\right|} \\
&\leq \frac{|w-2w|\left(\prod_{j=1}^{d_2}|w-cj^2w|\right)\left(\prod_{j=1}^{d_1}\left(w-\left\lfloor\frac{w}{cj^2}\right\rfloor\right)\right)}{\left(w-\frac{w}{c\cdot i^2}\right)\cdot\left(2w-\frac{w}{c\cdot i^2}\right)\left(\prod_{j=1}^{d_2}\left(w\cdot c\cdot j^2-\frac{w}{c\cdot i^2}\right)\right)\prod_{j=1,j\neq i}^{d_1}\left|\left\lfloor\frac{w}{c\cdot i^2}\right\rfloor-\left\lfloor\frac{w}{c\cdot j^2}\right\rfloor\right|} \\
&\leq \frac{|w-2w|\left(\prod_{j=1}^{d_2}|w-cj^2w|\right)\left(\prod_{j=1}^{d_1}\left(w-\frac{w}{cj^2}\right)\right)\cdot(1+o(1))}{\left(w-\frac{w}{c\cdot i^2}\right)\cdot\left(2w-\frac{w}{c\cdot i^2}\right)\left(\prod_{j=1}^{d_2}\left(w\cdot c\cdot j^2-\frac{w}{c\cdot i^2}\right)\right)\prod_{j=1,j\neq i}^{d_1}\left|\left\lfloor\frac{w}{c\cdot i^2}\right\rfloor-\left\lfloor\frac{w}{c\cdot j^2}\right\rfloor\right|}
\end{aligned} \tag{62}$$

Here, the final inequality used Equation (60). Let us consider the expression $\prod_{j=1,j\neq i}^{d_1}\left|\left\lfloor\frac{w}{c\cdot i^2}\right\rfloor - \left\lfloor\frac{w}{c\cdot j^2}\right\rfloor\right|$. This quantity is *at least*

$$\begin{aligned}
\prod_{j=1,j\neq i}^{d_1}\left(\left|\frac{w}{c\cdot i^2}-\frac{w}{c\cdot j^2}\right|-1\right) &= w^{d_1-1}\cdot\prod_{j=1,j\neq i}^{d_1}\frac{\left|j^2-i^2\right|-\frac{ci^2j^2}{w}}{ci^2j^2} \\
&= w^{d_1-1}\cdot\prod_{j=1,j\neq i}^{d_1}\frac{|j-i|\cdot|j+i|-\frac{ci^2j^2}{w}}{ci^2j^2}
\end{aligned}$$

$$= \left(\frac{w}{ci^2}\right)^{d_1-1} \cdot \prod_{j=1, j\neq i}^{d_1} \frac{|j-i| \cdot |j+i| - \frac{ci^2j^2}{w}}{j^2} \tag{63}$$

We claim that Expression (63) is at least

$$\left(\frac{w}{ci^2}\right)^{d_1-1} \cdot \frac{1}{2}. \tag{64}$$

In the case that $c = 2$ and d_1 is (at most) $w^{1/3}$, this is precisely [57, Claim 4]. We will ultimately take c to be a constant strictly greater than 2 and hence $d_1 = \lfloor (w/c)^{1/3} \rfloor$ is a constant factor smaller than $w^{1/3}$. The proof of [57, Claim 4] works with cosmetic changes in this case. For completeness, we present a derivation of the claim in Section A.

Equation (64) implies that Expression (62) is at most:

$$\begin{aligned}
&\frac{|w-2w| \left(\prod_{j=1}^{d_2} |w - cj^2 w|\right) \left(\prod_{j=1}^{d_1} \left(w - \frac{w}{cj^2}\right)\right) \cdot (1+o(1))}{\left(w - \frac{w}{c\cdot i^2}\right) \cdot \left(2w - \frac{w}{c\cdot i^2}\right) \left(\prod_{j=1}^{d_2} \left(w \cdot c \cdot j^2 - \frac{w}{c\cdot i^2}\right)\right) \left(\frac{w}{ci^2}\right)^{d_1-1} \cdot \frac{1}{2}} \\
&= \frac{2 \left(\prod_{j=1}^{d_2} |1 - cj^2|\right) \left(\prod_{j=1}^{d_1} \left(1 - \frac{1}{cj^2}\right)\right) \cdot (1+o(1))}{\left(1 - \frac{1}{c\cdot i^2}\right) \cdot \left(2 - \frac{1}{c\cdot i^2}\right) \left(\prod_{j=1}^{d_2} \left(c \cdot j^2 - \frac{1}{c\cdot i^2}\right)\right) \left(\frac{1}{ci^2}\right)^{d_1-1}} \\
&= \frac{2 \left(\prod_{j=1}^{d_2} (j^2 - 1/c)\right) \left(\prod_{j=1}^{d_1} \left(1 - \frac{1}{cj^2}\right)\right) \cdot (1+o(1))}{\left(1 - \frac{1}{c\cdot i^2}\right) \cdot \left(2 - \frac{1}{c\cdot i^2}\right) \left(\prod_{j=1}^{d_2} \left(j^2 - \frac{1}{c^2\cdot i^2}\right)\right) \left(\frac{1}{ci^2}\right)^{d_1-1}} \\
&\leq \frac{2\,(1+o(1))}{\left(1 - \frac{1}{c\cdot i^2}\right) \cdot \left(2 - \frac{1}{c\cdot i^2}\right) \left(\frac{1}{ci^2}\right)^{d_1-1}} \leq 4 \cdot \left(ci^2\right)^{d_1-1}.
\end{aligned} \tag{65}$$

Summarizing Equations (61) and (65), we have shown that: for any quantity $c \cdot i^2 \cdot w \in T_2$,

$$\left|\Phi(c \cdot w \cdot i^2)\right| / \left|\Phi(w)\right| \leq \frac{2\,(1+o(1))}{\left(1 - \frac{1}{c\cdot i^2}\right) \cdot (c \cdot i^2 - 2) \cdot (c \cdot i^2)^{d_1}} \tag{66}$$

and for any quantity $\left\lfloor \frac{w}{c\cdot i^2} \right\rfloor \in T_1$,

$$\left|\Phi\left(\left\lfloor \frac{w}{c \cdot i^2} \right\rfloor\right)\right| / \left|\Phi(w)\right| \leq 4 \cdot \left(ci^2\right)^{d_1-1}. \tag{67}$$

Let $\phi = \Phi/C$, where C is as in Equation (49). Let $D_1 = d_1$ and $D_2 = d_2$. Lemma 28 implies that ϕ is a feasible solution for the dual linear program of Section 4.4.1. We now show that, for any constant $\delta > 0$, by choosing c to be a sufficiently large constant (that depends on δ), we can ensure that ϕ achieves objective value $1 - 2\delta$.

Let

$$\begin{aligned}
A &= |\Phi(w)| \cdot w^{D_1}, \\
B &= |\Phi(2w)| \cdot (2w)^{D_1},
\end{aligned}$$

and

$$E = \sum_{i=1}^{d_1} |\Phi(\lfloor w/ci^2 \rfloor)| \cdot \left(\lfloor w/ci^2 \rfloor\right)^{D_1} + \sum_{i=1}^{d_2} |\Phi(\lfloor w \cdot ci^2 \rfloor)| \cdot \left(w \cdot c \cdot i^2\right)^{D_1}.$$

By Equation (58), $C = A + B + E$.

Moreover, observe that $\mathrm{sgn}(\Phi(w)) = -\mathrm{sgn}(\Phi(2w))$, so without loss of generality we may assume $\Phi(w) \geq 0$ and $\Phi(2w) \leq 0$ (if not, then replace Φ with $-\Phi$ throughout).

We now claim that by choosing c to be a sufficiently large constant, we can ensure that $E \leq \delta \cdot A$. To see this, observe that Equations (66) and (67), along with the fact that $D_1 = d_1$ and $D_2 = d_2$ implies that

$$\begin{aligned} E/A &\leq \frac{1}{w^{D_1}} \left[\left(\sum_{i=1}^{d_1} \left(\lfloor w/ci^2 \rfloor\right)^{D_1} \cdot 4 \cdot \left(ci^2\right)^{d_1 - 1} \right) + \left(\sum_{i=1}^{d_2} \left(w \cdot c \cdot i^2\right)^{D_1} \frac{2\left(1 - \frac{1}{c \cdot i^2}\right)(1 + o(1))}{(c \cdot i^2 - 2) \cdot (c \cdot i^2)^{d_1}} \right) \right] \\ &\leq \frac{1}{w^{D_1}} \left[\left(\sum_{i=1}^{d_1} \left(w/ci^2\right)^{D_1} \cdot 4 \cdot \left(ci^2\right)^{d_1 - 1} \right) + \left(\sum_{i=1}^{d_2} \left(w \cdot c \cdot i^2\right)^{D_1} \frac{2\left(1 - \frac{1}{c \cdot i^2}\right)(1 + o(1))}{(c \cdot i^2 - 2) \cdot (c \cdot i^2)^{d_1}} \right) \right] \\ &\leq 4 \left(\sum_{i=1}^{d_1} \frac{1}{c \cdot i^2} \right) + \left(\sum_{i=1}^{d_2} \frac{2(1 + o(1))}{\left(1 - \frac{1}{c \cdot i^2}\right)(c \cdot i^2 - 2)} \right) \end{aligned}$$

Since $\sum_{i=1}^{\infty} 1/(ci^2) \leq \frac{\pi^2}{6c}$, we see that choosing c to be a sufficiently large constant depending on δ ensures that $E/A \leq \delta$ as desired.

Hence, ϕ achieves objective value at least

$$\begin{aligned} \phi(w) \cdot w^{D_1} - \phi(2w) \cdot (2w)^{D_1} - \sum_{\ell \in \{1,\ldots,N\}, \ell \notin \{w, 2w\}} |\phi(\ell)| \cdot \ell^{D_1} \\ \geq \frac{A + B - E}{A + B + E} \geq \frac{(1 - \delta)A + B}{(1 + \delta)A + B} \geq 1 - 2\delta. \end{aligned}$$

4.6 Extending the lower bound to QSampling unitarily

So far in this section we have proved upper and lower bounds on the power of quantum algorithms for approximate counting that have access to two resources (in addition to membership queries): copies of $|S\rangle$, and the unitary transformation that reflects about $|S\rangle$. The assumption of access to the reflection unitary is justified by the argument that, if we had access to a unitary that prepared $|S\rangle$, then it could be used to reflect about $|S\rangle$ as well.

Giving the algorithm access to just the two resources above is an appealing model to use for upper bounds, since it does not assume anything about the method by which copies of $|S\rangle$ are prepared. This means algorithms derived in this model work in many different settings. For example, the algorithm may be able to QSample because someone else simply handed the algorithm copies of $|S\rangle$, or perhaps several copies of $|S\rangle$ just happen to be stored in the algorithm's quantum memory as a side effect of the execution of some earlier quantum algorithm. The upper bound given in Theorem 5 applies in any of these settings.

On the other hand, since only permitting access to QSamples and reflections about $|S\rangle$ ties the algorithm's hands, lower bounds for this model (e.g., Theorem 4) could be viewed as weaker than is desirable. In particular, our original justification for allowing access to reflections about $|S\rangle$ was that access to a unitary that prepared the state $|S\rangle$ would in particular allow such reflections to be done. Given this justification, it is very natural to wonder whether our lower bounds extend beyond just QSamples and reflections, to algorithms that are given access to *some* unitary process that permits both QSampling and reflections about $|S\rangle$.

Note that an algorithm with access to such a unitary could potentially exploit the unitary in ways other than QSamples and reflections to learn information about $|S\rangle$. For example, the algorithm could choose to run the unitary on inputs that do not produce $|S\rangle$. More generally, given a quantum circuit that implements a unitary, it is possible to construct, in a completely black-box manner, the inverse of this unitary, and also a controlled version of the unitary. The algorithm may choose to run the inverse on a state other than $|S\rangle$ to learn some additional information that is not captured by access to QSamples and reflections alone.

In summary, in this section we ask whether we can we extend the lower bound of Theorem 4 to work in a model where the algorithm is given access to some unitary operator that conveys the power to both QSample and reflect about $|S\rangle$.[3] Via Theorem 31 below, we explain that the answer is yes.

It may seem convenient to assume that the unitary transformation preparing $|S\rangle$ maps the all-zeros state to $|S\rangle$. But this is not the most general method of preparing $|S\rangle$ by a unitary. A unitary U that maps the all-zeros state to $|S\rangle|\psi\rangle$ would also suffice to create copies of $|S\rangle$, since the register containing $|\psi\rangle$ can simply be ignored for the remainder of the computation. More formally, assume U behaves as

$$U|0^m\rangle = |S\rangle|\psi\rangle, \tag{68}$$

where $|S\rangle|\psi\rangle$ is some m-qubit state. Clearly we can use U to create as many copies of $|S\rangle$ as we like, which as a by-product also creates copies of $|\psi\rangle$. This unitary also lets us reflect about $|S\rangle$. To see how, first use this unitary to create a copy of $|\psi\rangle$, and then consider the action of the unitary $U(\mathbf{1}-2|0^m\rangle\langle 0^m|)U^\dagger$ on the state $|\phi\rangle|\psi\rangle$ for any state $|\phi\rangle$. We claim that this unitary acts as a reflection about $|S\rangle$ when restricted to the first register. This establishes that any U of this form subsumes the power of both QSamples and reflections about $|S\rangle$.

Let us also assume without loss of generality that $|S\rangle|\psi\rangle$ is orthogonal to $|0^m\rangle$ from now on. This can be achieved by adding an additional qubit to the input that is always negated by the unitary. That is, we could instead consider the map $(U \otimes X)|0^m\rangle|0\rangle = |S\rangle|\psi\rangle|1\rangle$, which is orthogonal to the starting state by construction, and only increases the value of m by 1.

Of course, the requirement that $U|0^m\rangle = |S\rangle|\psi\rangle$ does not fully specify U, as it does not prescribe how U behaves on other input states. A reasonable prescription is that U should behave "trivially" on other input states, so that it does not leak information about S by its behavior on other states. In tension with this prescription is the fact the rest of the unitary must depend on S, since the first column of the unitary contains $|S\rangle$, and the rest of the columns have to be orthogonal to this.

[3] We thank Alexander Belov (personal communication) for raising this question.

Alexander Belov (personal communication) brought to our attention a very simple construction of such a unitary that leaks minimal additional information about S. Consider the unitary U that satisfies $U|0^m\rangle = |S\rangle|\psi\rangle$ and $U|S\rangle|\psi\rangle = |0^m\rangle$, with U acting as identity outside $\mathrm{span}\{|0^m\rangle, |S\rangle|\psi\rangle\}$. U is simply a reflection about the state $\frac{1}{\sqrt{2}}\big(|0^m\rangle - |S\rangle|\psi\rangle\big)$. This state is correctly normalized because we assumed that $|S\rangle|\psi\rangle$ is orthogonal to $|0^m\rangle$. Clearly U is now fully specified on the entire domain (once we have fixed $|\psi\rangle$) and it does not seem to leak any additional information about S.

In order to prove concrete lower bounds on the cost of algorithms for approximate counting given access to U, we need to fix $|\psi\rangle$. To answer the question posed in this section, we only need to establish that there exists *some* choice of $|\psi\rangle$ for which our algorithms cannot be improved. (Note that we cannot hope to establish lower bounds for arbitrary $|\psi\rangle$, since $|\psi\rangle$ could just contain the answer to the problem we are solving.)

To this end we make the specific choice of $|\psi\rangle = |S\rangle$ and consider the unitary V that acts as the unitary U above with $|\psi\rangle = |S\rangle$. In other words, V maps $|0^m\rangle$ to $|S\rangle|S\rangle$, $|S\rangle|S\rangle$ to $|0^m\rangle$, and acts as identity on the rest of the space. We also assume that $|0^m\rangle$ is orthogonal to $|S\rangle|S\rangle$. In other words, V simply reflects about the state $\frac{1}{\sqrt{2}}\big(|0^m\rangle - |S\rangle|S\rangle\big)$.

As previously discussed, granting an algorithm access to this unitary V lends the algorithm at least as much power the ability to QSample and perform reflections about $|S\rangle$. How efficiently can we solve approximate counting with membership queries and uses of the unitary V?

We can use our Laurent polynomial method to establish optimal lower bounds in this model as well and we obtain lower bounds identical to Theorem 4.

Theorem 31. *Let Q be a quantum algorithm that makes T queries to the membership oracle for S, and makes R uses of the unitary V defined above (and its inverse and controlled-V). If Q decides whether $|S| = w$ or $|S| = 2w$ with high probability, promised that one of those is the case, then either*

$$T = \Omega\left(\sqrt{\frac{N}{w}}\right) \qquad or \qquad R = \Omega\left(\min\left\{w^{1/3}, \sqrt{\frac{N}{w}}\right\}\right). \tag{69}$$

Proof. We follow the same strategy as in the proof of Theorem 4. Recall that $x \in \{0,1\}^N$ denotes the indicator vector of the set S. We only need to show that such a quantum algorithm gives rise to a Laurent polynomial in $|S| := \sum_{i=1}^n x_i$, with maximum exponent $O(T+R)$ and minimum exponent at least $-O(R)$ (as shown in Lemma 21 for the QSamples and reflections model).

We can prove this exactly the same way as Lemma 21 is established. Our quantum algorithm starts out from a canonical starting state that does not depend on the input and hence each entry of the starting state is a degree-0 polynomial. Membership queries involve multiplication with an oracle whose entries are ordinary polynomials of degree at most 1. The only thing that remains is understanding what the entries of the unitary V look like. We claim that the entries of V are given by a polynomial of degree at most 2 in the entries of the input x, with all coefficients of this degree-2 polynomial equal to either a constant, or a constant multiple of $|S|^{-1}$.

To see this, note that V is simply a reflection about the state

$$\frac{1}{\sqrt{2}}\left(|0^m\rangle - |S\rangle|S\rangle\right) = \frac{1}{\sqrt{2}}\left(|0^m\rangle - \frac{1}{|S|}\left(\sum_i x_i|i\rangle\right)\left(\sum_j x_j|j\rangle\right)\right). \tag{70}$$

The coefficient in front of $|0^m\rangle$ is a degree-0 polynomial and the other nonzero coefficients are a polynomial of degree at most 2 in the entries of the input x, with each coefficient of this polynomial equal to a constant multiple of $|S|^{-1}$.

Hence, each entry of the unitary V is also a polynomial of degree at most 2 in the entries of the input x, with each coefficient of this degree-2 polynomial equal to either a constant, or a constant multiple of $|S|^{-1}$. The same also holds for controlled-V, since that unitary is just the direct sum of identity with V. V is also self-inverse, so we do not need to account for that separately.

After the algorithm has made all the membership queries and uses of V, each amplitude of the final quantum state can be expressed as a polynomial of degree $O(T+R)$ in the input x, in which all coefficients are constant multiples of $|S|^{-R}$. The acceptance probability $p(x)$ of this algorithm will be a sum of squares of such polynomials. Exactly as in the proof of Theorem 4, Lemma 11 implies that there is a univariate polynomial q of degree at most $O(T+R)$, with coefficients that are multiples of the coefficients of p, such that for all integers $k \in \{0,\ldots,N\}$,

$$q(k) := \mathbb{E}_{|X|=k}\left[p(X)\right]. \tag{71}$$

Since the coefficients of $p(X)$ are constant multiples of $|X|^{-2R}$, q is in fact a real Laurent polynomial in k, with maximum exponent at most $O(R+T)$ and minimum exponent at least $-2R$. The theorem follows by a direct application Theorem 25 to q. □

A Establishing Equation 64

A.1 A clean calculation establishing a loose version of equation 64

For clarity of exposition, we begin by presenting a relatively clean calculation that establishes a slightly loose version of Equation (64). Using just this looser bound, we would be able to establish that Equation (64) holds (with the constant 1/2 replaced by a slightly smaller constant) so long as we set d_1 to be $\Theta\left(w^{1/3}/\log w\right)$. A slightly more involved calculation (cf. Section A.2) is required to establish Equation (64) for our desired value of $d_1 = \lfloor (w/c)^{1/3} \rfloor$.

Expression (63) equals

$$\begin{aligned}
&\left(\frac{w}{ci^2}\right)^{d_1-1} \cdot \frac{i^2}{\left((d_1)!\right)^2} \cdot \prod_{j=1,j\neq i}^{d_1}\left(|j-i|\cdot|j+i| - \frac{ci^2j^2}{w}\right) \\
&= \left(\frac{w}{ci^2}\right)^{d_1-1} \cdot \frac{i^2}{\left((d_1)!\right)^2} \cdot \prod_{j=1,j\neq i}^{d_1}\left(|j-i|\cdot|j+i|\right)\cdot\left(1 - \frac{ci^2j^2}{w\cdot|j-i||j+i|}\right) \\
&= \left(\frac{w}{ci^2}\right)^{d_1-1} \cdot \frac{(d_1+i)!(d_1-i)!}{2\left((d_1)!\right)^2} \cdot \prod_{j=1,j\neq i}^{d_1}\left(1 - \frac{ci^2j^2}{w\cdot|j-i||j+i|}\right)
\end{aligned} \tag{72}$$

$$\geq \left(\frac{w}{ci^2}\right)^{d_1-1} \cdot \frac{1}{2} \cdot \prod_{j=1,j\neq i}^{d_1} \left(1 - \frac{ci^2j^2}{w \cdot |j-i||j+i|}\right)$$

$$\geq \left(\frac{w}{ci^2}\right)^{d_1-1} \cdot \frac{1}{2} \cdot \left(1 - \sum_{j=1,j\neq i}^{d_1} \frac{ci^2j^2}{w \cdot |j-i||j+i|}\right)$$

$$\geq \left(\frac{w}{ci^2}\right)^{d_1-1} \cdot \frac{1}{2} \cdot \left(1 - \frac{ci^2}{w} \sum_{j=1,j\neq i}^{d_1} \frac{j^2}{|j-i||j+i|}\right). \quad (73)$$

Let us consider the expression $\sum_{j=1,j\neq i}^{d_1} \frac{j^2}{|j-i||j+i|}$. If $i^2 \notin [j^2/2, 3j^2/2]$, then the j'th term in this sum is at most 2. Hence, letting H_i denote the ith Harmonic number and using the fact that $H_i \leq \ln(i+1)$,

$$\sum_{j=1,j\neq i}^{d_1} \frac{j^2}{|j-i||j+i|}$$

$$\leq 2 \cdot d_1 + \sum_{j=\lfloor\sqrt{2/3}i\rfloor}^{\lfloor\sqrt{2}i\rfloor} \frac{j^2}{|j-i||j+i|}$$

$$\leq 2 \cdot d_1 + \sum_{j=\lfloor\sqrt{2/3}\cdot i\rfloor}^{\lceil\sqrt{2}\cdot i\rceil} \frac{j}{|j-i|}$$

$$\leq 2d_1 + \sqrt{2} \cdot i \cdot \sum_{j=1}^{(\sqrt{2}-1)\cdot i} 2/j$$

$$\leq 2d_1 + 2\sqrt{2} \cdot i \cdot H_i \leq 2d_1 + 2\sqrt{2}i\ln(i+1). \quad (74)$$

We conclude that if d_1 were set to a value less than $w^{1/3}/(100 \cdot c^2 \cdot \ln(w))$ (rather than to $\lfloor(w/c)^{1/3}\rfloor$), then Expression (73) is at least

$$\left(\frac{w}{ci^2}\right)^{d_1-1} \cdot \frac{1-1/c}{2}. \quad (75)$$

A.2 The tight bound

To obtain the tight bound, we need a tighter sequence of inequalities following Expression (72). Specifically, Expression (72) is bounded below by:

$$\geq \left(\frac{w}{ci^2}\right)^{d_1-1} \cdot \frac{1}{2}\left(1 + \frac{i}{2d_1}\right)^i \cdot \prod_{j=1,j\neq i}^{d_1} \left(1 - \frac{ci^2j^2}{w \cdot |j-i||j+i|}\right)$$

$$\geq \left(\frac{w}{ci^2}\right)^{d_1-1} \cdot \frac{1}{2} \cdot e^{i^2/(2d_1)} \cdot \prod_{j=1,j\neq i}^{d_1} \left(1 - \frac{ci^2j^2}{w \cdot |j-i||j+i|}\right)$$

$$\geq \left(\frac{w}{ci^2}\right)^{d_1-1} \cdot \frac{1}{2} \cdot e^{i^2/(2d_1)} \cdot \prod_{j=1,j\neq i}^{d_1} \left(1 - \frac{ci^2j^2}{w \cdot |j-i||j+i|}\right) \quad (76)$$

The rough idea of how to proceed is as follows. Equation (74) implies that for $i \ll w^{1/3}/\ln w$, the factor

$$F_1 := \prod_{j=1, j\neq i}^{d_1} \left(1 - \frac{ci^2j^2}{w \cdot |j-i||j+i|}\right)$$

is at some a positive constant, and hence Expression (76) is bounded below by the desired quantity. If $i \gtrsim w^{1/3}/\ln w$, then Equation (74) does not yield a good bound on this factor, leaving open the possibility that this factor is subconstant. But in this case, the factor $F_2 := e^{i^2/(2d_1)} \geq e^{\tilde{\Omega}(d_1)}$, and the largeness of F_2 dominates the smallness of F_1.

In more detail, let $x_{i,j} = \frac{ci^2j^2}{w \cdot |i-j|j+i||}$. Then for all $i \neq j$ such that $i, j \leq d_1$,

$$x_{i,j} \leq \frac{c \cdot d_1^2(d_1-1)^2}{(2d_1-1) \cdot w} \leq \frac{c \cdot d_1^3}{2w} \leq 1/2, \tag{77}$$

where in the final inequality we used the fact that $d_1 \leq (w/c)^{1/3}$.

Using the fact that $1 - x \geq e^{-x-x^2}$ for all $x \in [0, 1/2]$, we can write

$$F_1 \geq \prod_{j=1, j\neq i}^{d_1} e^{-x_{i,j}-x_{i,j}^2}.$$

Hence,

$$F_1 \cdot F_2 \geq \exp\left(i^2/(2d_1) - \sum_{j=1, j\neq i}^{d_1} -x_{i,j} - x_{i,j}^2\right).$$

From Equations (74) and (77), we know that

$$\sum_{j=1, j\neq i}^{d_1} x_{i,j} + x_{i,j}^2 \leq \frac{ci^2}{w} \cdot \left(3d_1 + 3\sqrt{2}i\ln(i+1)\right) \leq \frac{ci^2}{w} \cdot (4d_1 \ln(d_1)).$$

Hence,

$$\begin{aligned} F_1 \cdot F_2 &\geq \exp\left(i^2/(2d_1) - \frac{ci^2}{w} \cdot 4c\ln(d_1)\right) \\ &= \exp\left(i^2\left(\frac{1}{2d_1} - \frac{4c^2\ln(d_1)}{w}\right)\right) \\ &\geq \exp\left(i^2 \cdot \frac{1}{2d_1} \cdot (1 - o(1))\right) \\ &\geq 1. \end{aligned}$$

Equation (64) follows.

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